

2.5 The Geometric Perspective on Matrix Multiplication



Core skills you'll master in this section:

- ▶ The geometric meaning of matrix multiplication: matrix multiplication is not just a numerical operation — it represents linear transformations such as rotation, scaling, projection, and shearing.
- ▶ $Ax = y$ transformation: matrix A maps the input vector x to the output vector y .
- ▶ 2×2 matrices: planar transformations, such as rotation, reflection, shearing, and projection.
- ▶ 3×3 matrices: rotation, scaling, shearing, and projection in 3D space.
- ▶ 3×2 matrices: a 2D vector is mapped into 3D space, but the result still lies within a lower-dimensional subspace.
- ▶ 2×3 matrices: a 3D vector is projected down into 2D space via the matrix.

A matrix — this table of numbers — can do a lot more than you might think! Beyond just storing numbers, a matrix's greater power comes from linear transformation. And the mathematical tool that a linear transformation relies on is precisely matrix multiplication.

Earlier in this book we gave matrix multiplication a first introduction. Please pay special attention: matrix multiplication is not a “numbers game,” and it's certainly not some clever trick!

Matrix multiplication is far more than arranging and operating on numbers — behind it lies a rich geometric meaning. From a geometric perspective, matrix multiplication is essentially a linear transformation, capable of carrying out operations such as rotation, scaling, shearing, and projection. Every matrix can be thought of as a “tool” for scaling, rotating, reflecting, projecting, or shearing a vector.

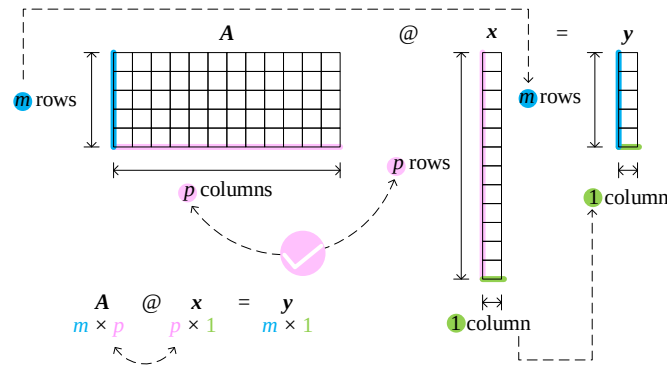
$$Ax = y$$

When we multiply a vector by a matrix, we are actually performing some geometric transformation on that vector.

To understand linear transformations from a geometric perspective, and thereby deepen our understanding of matrix multiplication, this chapter will start from the following matrix multiplication

$$A_{m \times p} x_{p \times 1} = y_{m \times 1}$$

Looking at the matrix shapes, $(m \times p) @ (p \times 1)$ — the (p) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape $(m \times 1)$.

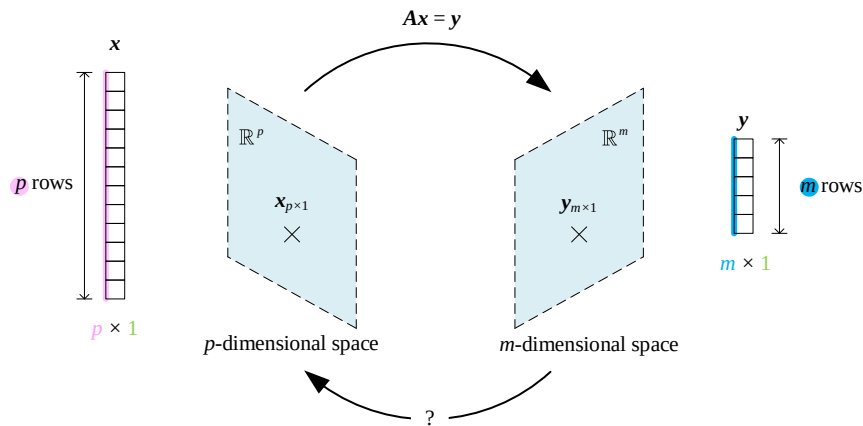
Figure 1. The Matrix Multiplication $Ax = y$

Linear Transformation

As shown in Figure 2, from the perspective of a linear mapping, $Ax = y$ represents the mapping of the p -dimensional column vector x under the action of A , producing an m -dimensional column vector y .

Column vector $x_{p \times 1}$ is in $\mathbb{R}^p$, while column vector $y_{m \times 1}$ is in $\mathbb{R}^m$.

That is, $Ax = y$ is a specific mapping from $\mathbb{R}^p$ space to $\mathbb{R}^m$ space.

Figure 2. The Linear Mapping $Ax = y$

The p -dimensional column vector x serves as the input vector, representing some initial state; we can think of x as a point in $\mathbb{R}^p$ space.

The m -dimensional column vector y serves as the output vector, representing the result after the transformation; we can think of x as a point in $\mathbb{R}^m$ space.

And matrix A determines the rule for the geometric transformation; this geometric transformation might be scaling, rotation, projection, reflection, or an ordered combination of these.

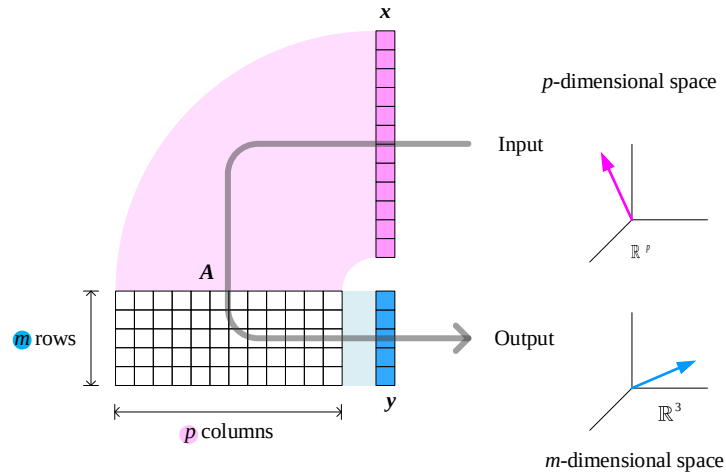
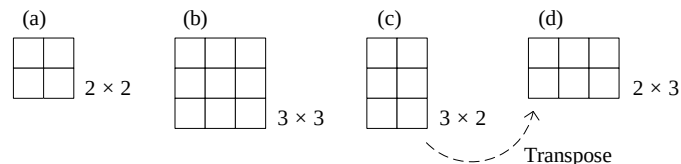
Figure 3. The Matrix Multiplication $Ax = y$'s Input and Output

Figure 3 — this way of arranging matrix multiplication is the first perspective on matrix multiplication, a topic we'll cover in the next chapter.

Four Examples

In this section, we'll take a deep dive into how matrix multiplication acts on vectors through geometric transformations.

Specifically, we'll analyze how four differently shaped matrices in Figure 4 — 2×2 , 3×3 , 3×2 , and 2×3 — affect vectors.

Figure 4. Four Different Matrices A

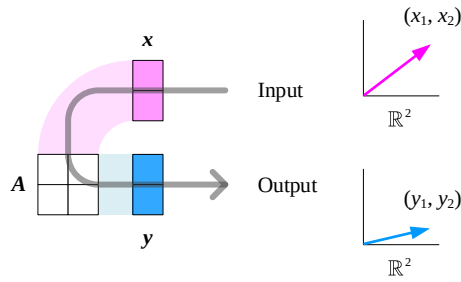
2×2 Matrices

2×2 matrices are commonly used for planar geometric operations in 2D space, such as scaling, rotation, reflection, projection, and shearing.

For the matrix multiplication $Ax = y$, when matrix A 's shape is 2×2 , the input vector x is a 2×1 vector; the output vector y is also a 2×1 vector.

Looking at the matrix shapes, $(2 \times 2) @ (2 \times 1)$ — the (2) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (2×1) .

As required by the rules of matrix multiplication, matrix A 's number of columns matches the input vector x 's number of rows; and matrix A 's number of rows matches the output vector y 's number of rows.

Figure 5. Matrix A is 2×2

From a geometric perspective, the 2D column vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ corresponds to a point in the planar Cartesian coordinate system $\mathbb{R}^2$, at (x_1, x_2) ; $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is also a vector in the plane, starting at the origin $(0,0)$ and ending at (x_1, x_2) .

And the 2D column vector $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ is likewise a point in the planar Cartesian coordinate system, at (y_1, y_2) ; $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ is also a vector in the planar Cartesian coordinate system $\mathbb{R}^2$, starting at the origin $(0,0)$ and ending at (y_1, y_2) .

As mentioned earlier in this book, $\mathbb{R}^2$ denotes the two-dimensional real vector space. Simply put, a vector space is a set whose elements (called vectors) support addition and scalar multiplication and satisfy certain axioms (such as associativity and distributivity), together forming a mathematical space with a linear structure.

Next, let's first look at two special vectors,

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

As mentioned earlier, these are the standard basis vectors of the planar Cartesian coordinate system; the unit vector $\mathbf{e}_1$ points in the positive x_1 direction, and the unit vector $\mathbf{e}_2$ points in the positive x_2 direction.

Given the following 2×2 matrix A

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The matrix multiplication $A @ \mathbf{e}_1$

$$A @ \mathbf{e}_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}$$

is equivalent to extracting matrix A 's 1st column. Similarly, $A @ \mathbf{e}_2$ extracts matrix A 's 2nd column.

As shown in figure (a) below, on the “red-green” plane, $\mathbf{e}_1$ represents the pure red vector, and $\mathbf{e}_2$ represents the pure green vector.

In figure (b) below, the regularly arranged square scatter points on the “red-green” plane are actually color vectors generated by linear combinations of $\mathbf{e}_1$ and $\mathbf{e}_2$.

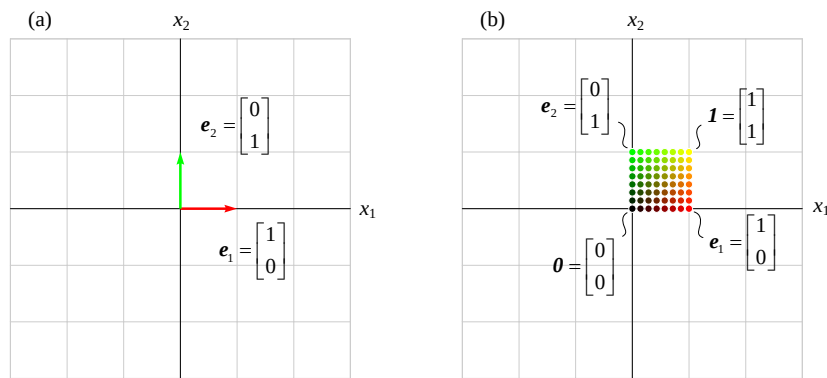


Figure 6. Unit Vectors and Regularly Arranged Scatter Points on the “Red-Green” Plane

Next, we'll use different 2×2 matrices A to map the standard basis vectors e_1 and e_2 on the “red-green” plane, along with the scatter points, and observe how they transform from the original square distribution into various different geometric shapes.

Please analyze the geometric transformation corresponding to each row of Table 1 one by one.

We can also observe that after the original square grid is mapped by matrix A , it becomes a grid of evenly spaced parallelograms. It's worth noting that no matter how matrix A changes, the origin's position always stays fixed — this is an important property of linear mappings.

Here, we just want you to form an initial, intuitive impression of linear transformations; later chapters of this book will dive deeper into the mathematical principles of linear mappings and their wide-ranging applications.

Also, readers who don't yet understand the principles behind these planar geometric transformations shouldn't worry — this book will cover this material in dedicated detail later on.

Table 1. Examples of Geometric Transformations Produced by the 2×2 Matrix A

Matrix	Matrix Multiplication (try computing by hand)	Before and After the Geometric Transformation
$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ Unchanged	$A @ e_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$	
$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ Uniformly Stretched 2x	$A @ e_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$	

$A = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$ Squeeze	$A @ e_1 = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ Mirrored About the x1 Axis	$A @ e_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$	
$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ Flipped Symmetrically About the Origin	$A @ e_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$	
$A = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix}$ Rotated 30° Counterclockwise About the Origin	$A @ e_1 = \begin{bmatrix} \cos 30^\circ \\ \sin 30^\circ \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} -\sin 30^\circ \\ \cos 30^\circ \end{bmatrix} = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ Sheared Along the x1 Direction	$A @ e_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	
$A = \begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{bmatrix}$ Mirrored About a Certain Line	$A @ e_1 = \begin{bmatrix} \cos 60^\circ \\ \sin 60^\circ \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$ $A @ e_2 = \begin{bmatrix} \sin 60^\circ \\ -\cos 60^\circ \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$	

3 × 3 Matrices

3 × 3 matrices extend into 3D space, capable of handling more complex spatial geometric transformations.

For the matrix multiplication $\mathbf{Ax} = \mathbf{y}$, when matrix $\mathbf{A}$'s shape is 3×3 (3 rows, 3 columns), the input vector $\mathbf{x}$ is a 3×1 vector, and $\mathbf{x}$ represents a coordinate point in 3D space $\mathbb{R}^3$; the output vector $\mathbf{y}$ is also a 3×1 vector, and $\mathbf{y}$ is likewise a coordinate point in 3D space $\mathbb{R}^3$.

The 3×3 matrix $\mathbf{A}$ carries out the mapping from the 3D vector $\mathbf{x}$ to the 3D vector $\mathbf{y}$.

Looking at the matrix shapes, $(3 \times 3) @ (3 \times 1)$ — the (3) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (3×1) .

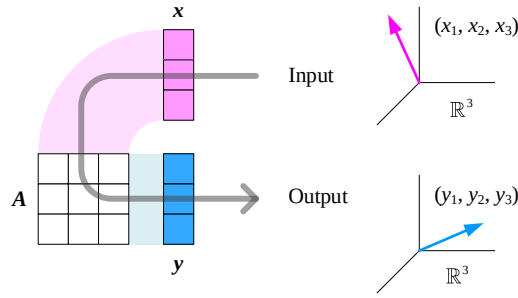


Figure 7. Matrix $\mathbf{A}$ is 3×3

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The 3D column vector $\mathbf{x}$ is a point in the 3D Cartesian coordinate system, at (x_1, x_2, x_3) ;

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is also a vector in the 3D Cartesian coordinate system $\mathbb{R}^3$, starting at the origin $(0,0,0)$ and ending at (x_1, x_2, x_3) .

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

The 3D column vector $\mathbf{y}$ represents a point in the 3D Cartesian coordinate system, at $(y_1, y_2,$

$y_3)$; $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ is likewise a vector in the 3D Cartesian coordinate system $\mathbb{R}^3$, starting at the origin $(0,0,0)$ and ending at (y_1, y_2, y_3) .

As shown in figure (a) below, in RGB color space, $\mathbf{e}_1$ represents the pure red vector $[1, 0, 0]^T$, $\mathbf{e}_2$ represents the pure green vector $[0, 1, 0]^T$, and $\mathbf{e}_3$ represents the pure blue vector $[0, 0, 1]^T$. In figure (b) below, the regularly arranged scatter points along the RGB cube's twelve edges are actually color vectors generated by linear combinations of $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$.

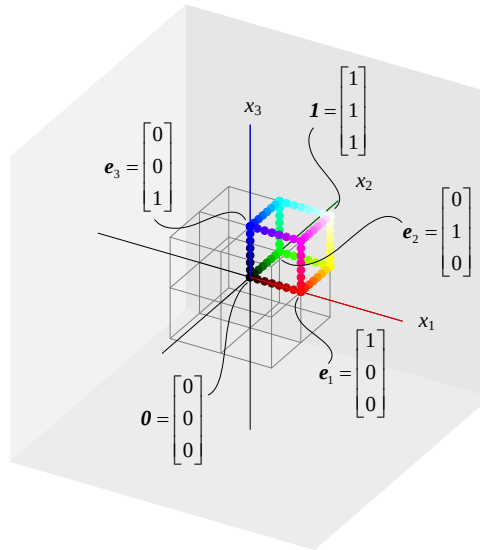


Figure 8. Unit Vectors and Regularly Arranged Scatter Points in “Red-Green-Blue” Space

Table 2 gives examples of linear transformations by 3×3 matrices – please analyze them one by one on your own.

Table 2. Examples of Geometric Transformations Produced by the 3×3 Matrix A

Matrix	Matrix Multiplication (try computing by hand)	Before and After the Geometric Transformation
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Unchanged	$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	
$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ Uniformly Stretched 2x	$Ae_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$	
$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 1.25 \end{bmatrix}$ Non-Uniform Scaling	$Ae_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 0.8 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 1.25 \end{bmatrix}$	

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{bmatrix}$ Rotated About the x_1 Axis	$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Mirrored About the x_1x_2 Plane	$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \tan 30^\circ & 0 & 1 \end{bmatrix}$ Sheared Along the x_3 Axis	$Ae_1 = \begin{bmatrix} 1 \\ 0 \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad Ae_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad Ae_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	

3 × 2 Matrices

3 × 2 and 2 × 3 matrices, meanwhile, involve conversions between different dimensions — for example, embedding a 2D vector into 3D space, or mapping a 3D vector down into 2D space.

As shown in Figure 9, if matrix A 's shape is 3 × 2, the input vector x is 2-dimensional — that is, a point in the plane; the output vector y is 3-dimensional, representing a point in 3D space.

Looking at the matrix shapes, for the matrix multiplication $Ax = y$, (3 × 2) @ (2 × 1) — the (2) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (3 × 1).

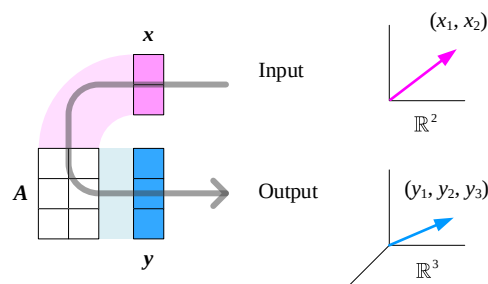


Figure 9. Matrix A is 3 × 2

Table 3 gives a few examples — please analyze them on your own.

The main point to note is that when we apply a 3 × 2 matrix A to a 2D vector x , the result is a 3D vector y , which seems to “raise the dimension” of the original 2D plane up into 3D space. However, a closer look at the transformed result shows that although the coordinates expanded from 2D to 3D, these points don't fill the entire 3D space — they still lie on a plane.

This is because the space spanned by the column vectors of the 3×2 matrix $\mathbf{A}$ in the table has a maximum dimension of a 2D plane; although this plane is embedded in 3D space, it still retains its 2D character. So this “dimension-raising” is really a kind of embedding, not a genuine expansion of dimension. Note that the dimension of the space spanned by the 3×2 matrix $\mathbf{A}$'s column vectors is not always 2D.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 0 \end{bmatrix}_s$$

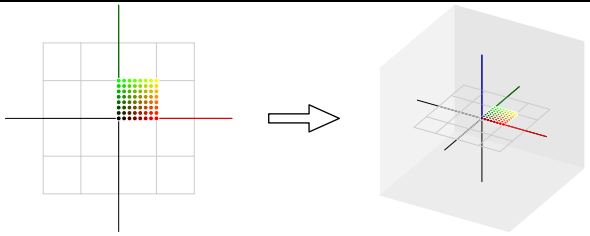
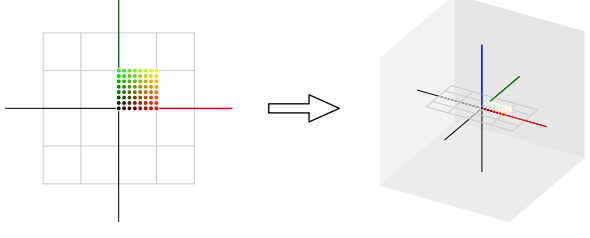
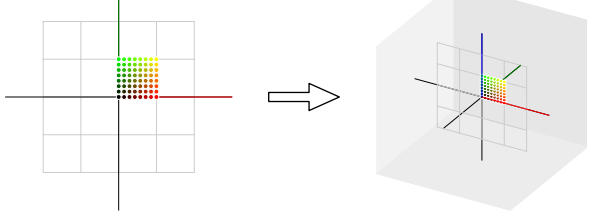
In some cases, $\mathbf{A}$'s column vectors can span a space of dimension 1. For example, if column vectors are linearly dependent (that is, one vector is a multiple of the other), then $\mathbf{A}$'s column vectors span a line in 3D space.

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

If $\mathbf{A}$'s column vectors are all **0** zero vectors — that is, — then $\mathbf{A}$'s column vectors span a space of dimension 0.

These cases show that the linear dependence of the 3×2 matrix $\mathbf{A}$'s column vectors determines the dimension of the transformed space. In later chapters, we'll further explore a matrix's rank, column space, and the geometric meaning of linear transformations, gaining a deeper understanding of the mathematics behind these phenomena. By studying these concepts, we can better grasp how to use matrices to describe and analyze geometric transformations, and how to infer the properties of the transformed space from a matrix's characteristics.

Table 3. Examples of Geometric Transformations Produced by the 3×2 Matrix $\mathbf{A}$

Matrix	Matrix Multiplication (try computing by hand)	Before and After the Geometric Transformation
$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ Dimension-Raising	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	
$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \\ 0 & 0 \end{bmatrix}$ Scaling + Dimension-Raising	$\begin{bmatrix} 1 & 0 \\ 0 & 0.5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0.5x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	
$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$ Raised into the x1x3 Plane	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	

$A = \begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$ Central Symmetry + Dimension-Raising	$\begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -x_1 \\ -x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 \\ 0 & \cos 60^\circ \\ 0 & \sin 60^\circ \end{bmatrix}$ Rotation + Dimension-Raising	$\begin{bmatrix} 1 & 0 \\ 0 & \cos 60^\circ \\ 0 & \sin 60^\circ \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ \cos 60^\circ x_2 \\ \sin 60^\circ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ Shear + Dimension-Raising	$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 + x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$	

2 × 3 Matrices

As shown in Figure 10, if matrix A 's shape is 2×3 , the input vector x is 3-dimensional — that is, a point in 3D space; the output vector y is 2-dimensional, representing a point in the plane.

Looking at the matrix shapes, $(2 \times 3) @ (3 \times 1)$ — the (3) sandwiched in the middle gets “canceled out,” so the result of the matrix multiplication has shape (2×1) .

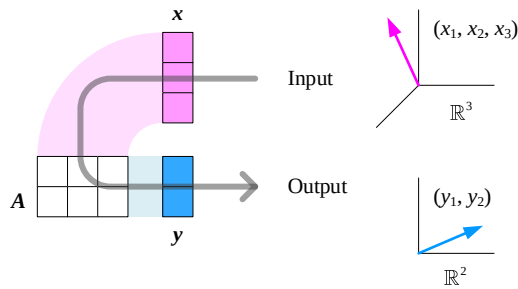


Figure 10. Matrix A is 2×3

From a linear-mapping perspective, the 2×3 flat matrix A carries out the mapping from the 3D column vector x to the 2D column vector y .

Please analyze the examples in Table 4 on your own.

When we apply a 2×3 matrix A to a 3D vector x , the result is a 2D vector y , which “reduces the dimension” of the original 3D space down to a 2D plane. This “dimension reduction” is really a kind of projection — it compresses information from 3D space into a lower-dimensional space while preserving certain linear structure.

Table 4. Examples of Geometric Transformations Produced by the 2×3 Matrix A

Matrix	Matrix Multiplication (try computing by hand)	Before and After the Geometric Transformation
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ Dimension-Reducing	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}$ Scaling + Dimension-Reducing	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 0.5x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
$A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Central Symmetry + Dimension-Reducing	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 \\ -x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60^\circ & \sin 60^\circ \end{bmatrix}$ Rotation + Dimension-Reducing	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60^\circ & \sin 60^\circ \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ \cos 60^\circ x_2 + \sin 60^\circ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ Shear + Dimension-Reducing	$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	
$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ Projection + Scaling + Dimension-Reducing	$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	

Inverse Transformation

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Code and PDF file downloads: <https://github.com/Visualize-ML>

The companion video lectures for this book are all published on Bilibili — 生姜 DrGinger: <https://space.bilibili.com/513194466>

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Figure 2 showed this mapping. But whether an “inverse transformation” exists — one that converts y back into x — depends on whether matrix A has a matrix inverse. A can only have an inverse if it is a square matrix.

The matrix inverse means that when square matrix A is invertible, there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I$$

Here, I is the identity matrix with the same shape as square matrix A .

That is, A^{-1} can reverse the transformation performed by A .

For example,

$$\underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}}_A @ \underbrace{\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}}_{A^{-1}} @ \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_I$$

As shown in Figure 11, matrix A scales everything up by a factor of 2, and the inverse matrix A^{-1} scales everything back down by a factor of $1/2$.

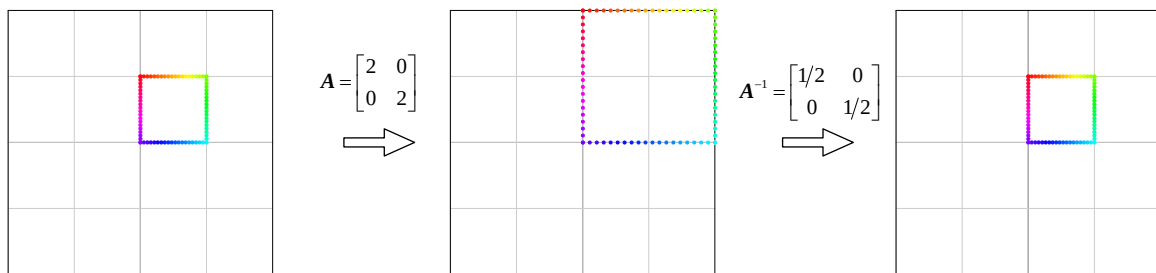


Figure 11. Uniform Scaling in the Plane, the 2×2 Matrix

Whether we scale up first (A) and then scale down (A^{-1}), or scale down first (A^{-1}) and then scale up (A), the planar geometric shape is restored either way, which corresponds to $AA^{-1} = A^{-1}A = I$.

➡ This is just a preview — this book will cover the topic of matrix inverses in dedicated detail later on.

Through these examples, we've revealed how matrix multiplication achieves a rich variety of geometric transformations across spaces of different dimensions. With these geometric perspectives in hand, understanding the properties of matrix multiplication becomes much easier — which is what we'll cover at the end of this chapter.

Beyond helping us understand matrix multiplication, these geometric perspectives will also play a crucial role in understanding various matrix decompositions.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. In the plane, how can matrix multiplication be used to perform scaling?

Q2. In 3D space, how can matrix multiplication be used to perform rotation about the x_1 , x_2 , and x_3 axes, respectively?

Q3. In the plane, how can matrix multiplication be used to perform rotation?

Q4. In 3D space, how can matrix multiplication be used to perform scaling?

Q5. In the plane, how can matrix multiplication be used to perform reflection about the x_1 axis?

Q6. In 3D space, how can matrix multiplication be used to perform reflection about the x_1x_2 plane?

Q7. In the plane, how can matrix multiplication be used to perform shearing along the x_1 axis?

Q8. In 3D space, how can matrix multiplication be used to perform shearing along the x_1 axis?